Elena Viscardi

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elena-viscardi-astrophysics 0009-0005-5187-6074 NASA ADS: Viscardi, E./ E. M.

Research Interests

Research question: How can planets form from the dust and gas in protoplanetary disks?
Methodology: Multi-wavelength radiative transfer modeling of continuum emission and kinematical modeling of gas line emission to measure key disk properties: disk mass and size, gas-to-dust ratio, dust grain size, dust composition, and turbulence. Comparison of these properties with planet formation simulations.
Data: Radio interferometric ALMA and VLA data + synergy with optical/infrared observations (VLT, JWST).

UAT Keywords: Protoplanetary disks; Radio interferometers; Bayesian statistics.

Employment

PhD Candidate in Physics *11/2023 – 06/2027*
Institute: European Southern Observatory (ESO) *Garching, Germany*
Thesis: Quantifying planetesimal formation in protoplanetary disk substructures
Supervisors: Dr. E. Macías (ESO), Prof. T. Birnstiel (LMU), Dr. F. Zagaria (MPIA)
Work contract extended by 7 months due to health issues

Education

MSc in Physics (110/110 cum laude) *09/2021 – 09/2023*
Institute: University of Milan (UniMi) *Milan, Italy*
Thesis: Measuring the radial extent of Chamaeleon protoplanetary discs in CO emissions
Supervisors: Prof. G. Rosotti (UniMi), Prof. G. Lodato (UniMi), Dr. C. Toci (UniMi)

BSc in Physics (110/110 cum laude) *10/2018 – 10/2021*
Institute: University of Milan (UniMi) *Milan, Italy*
Thesis: Disc mass measurements in WaOph6 from ^{12}CO emission with DISCMINER
Supervisors: Prof. G. Lodato (UniMi), Dr. A. Izquierdo (ESO), Dr. T. Paneque (ESO)

Awards

Accepted as a member of the International Max Planck Research School (IMPRS, one of 22 out of 437 applicants). *Garching, Germany. 03/2023.*

Observing Experience

Facilities: ALMA and VLA (radio interferometers); VLT/MUSE (integral-field optical spectrograph); VLT/UVES (high-resolution optical spectrograph).
Accepted Proposals: 9 accepted proposals as CoI across ALMA and VLA. Outcomes awaited for 3 VLA proposals (one as PI) and 2 VLT proposals (one as PI).
Proposal writing: Submitted 10 observing proposals as PI (6 ALMA, 1 VLA, 2 VLT/MUSE NFM, 1 VLT/UVES), developing the scientific case and designing the spectral setups.
Proposal review: 80 proposals reviewed across ALMA, VLA, and VLT.

Data Analysis & Computing

Data reduction: Archive mining, calibration, imaging, and self-calibration of ALMA and VLA data.
Data analysis: Analysis of continuum and spectral-line radio interferometric data across multiple frequency bands and angular resolutions. Visibility modeling (`frank`, `galario`); kinematical analysis (`bettermoments`, `discminer`); multi-wavelength radiative transfer modeling (own code, available on GitHub).
Simulations: Dust growth and coagulation with `dustpy`; simulated observations and observing setups for current and future facilities (e.g., ngVLA) using radiative transfer and CASA `simobserve`.

Mentoring

Co-supervisor & Organizer, ESO Summer Research Project *06–08/2025*
Co-organizer of the ESO Summer Research Project. Co-supervised a student (Paula Manuela Pineda), providing guidance in data reduction, visibility modeling, theoretical background, and scientific presentation skills.

Co-supervisor, Master's Thesis, University of Milan *03–09/2024*
Co-supervised a Master's student (Stefano Gelpi), providing guidance in imaging, geometrical analysis, theoretical background, and scientific presentation skills.

Talks, Scientific Visits & Collaborations

I collaborate with researchers from 25+ host institutions across Europe, Asia, Australia, North and South America (map).

Contributed talks

09/2026	Bridging it (Heidelberg)
07/2026	Disc on the Exe (Exeter)
11/2025	Dusty Universe 2025 (Tucson)
09/2025	ORIGINS Workshop "From Disk Formation to Exoplanetary Systems" (Garching)
09/2025	IMPRS Symposium (Garching)
02/2025	Pebbles in Planet Formation (Tokyo)
12/2024	Milan Christmas Workshop (Milan)
11/2024	ngVLA Workshop "Follow the Monarchs" (Morelia)
06/2024	EAS Annual Meeting, SS5 and SS19 (Padua)
12/2023	Milan Christmas Workshop (Milan)
12/2021	Milan Christmas Workshop (Milan)

Scientific visits

04/2026	EARLYBIRD Group Science Retreat, LMU Munich (Chiemsee)
03/2026	Planet Formation & Exoplanets Group, MPIA (Heidelberg)
01/2026	Stellar-MADE Group, IPAG, Univ. of Grenoble (Grenoble)
11/2024	Institute of Radioastronomy & Astrophysics, UNAM (Morelia)
04–06/2024	Star & Planet Formation Group, Univ. de Chile (Santiago)
12/2023	Star & Planet Formation Group, Univ. of Milan (Milan)

Science Community work & Leadership

Scientific Assistant, ESO/VLT OPC Panel Meetings (P118) *11/2026*
Host and facilitator for the expert panel meetings reviewing VLT observing proposals; handling conflicts of interest and ensuring adherence to the Code of Conduct.

Member, ESO Studentships Selection Committee *10/2024 – 04/2025*
Evaluated application materials and helped conduct interviews, supporting an equitable, transparent, and unbiased selection process.

Cordinator, IMPRS Advanced Course *03/2025*
Coordinated with IMPRS a one-week doctoral course, "Topics in Star and Planet Formation" by visiting professors Nuria Calvet and Lee Hartmann, attended by 20+ students.

Local Organizing Committee & Technical Support *07/2024 – 03/2025*
Two international conferences (*Towards New Frontiers* and *New Heights in Planet Formation*): managed logistics, provided technical support, and served as Code of Conduct representative.

Organizing Committee, ESO Journal Club & Informal Discussions *09/2024 – Present*
Organizer of two weekly ESO talk series: the Journal Club (current) and the Informal Discussions (09/2024 – 09/2025). Schedule sessions, recruit speakers, and chair meetings.

EDI Student Representative, Office for Science *11/2024 – 11/2025*
Advocated for extending EDI representation to students in Engineering and Science Communication departments. Drafted a "Mutual Expectations" document for facilitating student–supervisor relationships. Co-initiated an EDI seminar series on workplace well-being, harassment prevention, and selection biases.

Organizer, Students & Fellows Mental Wellbeing Event *05/2025*
Organized and hosted a workshop for 25+ early-career researchers on burnout, stress, and impostor syndrome. Invited the ESO Ombuds to speak on conflict resolution and healthy workplace communication

Outreach

Co-organizer of **Astronomy on Tap Munich**, a monthly "after-work" public event with two astronomy talks, a pub quiz, and a news round-up.

Skills

Programming: Python (advanced), C/C++ (intermediate).

Software: CASA, CARTA, GALARIO, frank, frap, emcee, UltraNest, DISCMINER, beam_boozle, gofish, eddy, bettermoments, optool, dustpy, dust_opac.

Languages: Italian (native), English (fluent).